

lm-eval-harness

평가 프레임워크



lm-eval-harness

- <https://github.com/EleutherAI/lm-evaluation-harness>
- Language Model Evaluation Harness
- Eleuther.ai가 주도하여 개발된 평가 프레임워크
 - Huggingface의 LLM leaderboard와 korean LLM leaderboard에서 평가 프레임워크로 채택하여 운영
- ARC(추론을 필요로 하는 질의응답), HellaSwag(상식 기반 추론), MMLU(57개 영역의 지식 테스트), 그리고 TruthfulQA(사실성 및 정확성 평가)와 같은 다양한 벤치마크를 통해 언어모델을 평가

LM-Evaluation-Harness

- 설치

```
git clone --depth 1 https://github.com/EleutherAI/lm-evaluation-harness
cd lm-evaluation-harness
pip3 install -e .
```

- 설치

```
pip3 install "lm_eval[hf]"
pip3 install "lm_eval[vllm]"
pip3 install "lm_eval[api]"
```

```
pip3 show lm_eval
```

LM-Evaluation-Harness(hellaswag)

- 태스크 목록 / 실행

```
lm-eval ls tasks
```

```
lm_eval --model hf \
  --model_args pretrained=meta-llama/Llama-3.2-1B-Instruct \
  --tasks hellaswag \
  --device cuda:0 \
  --batch_size 2
```

Tasks	Version	Filter	n-shot	Metric		Value		Stderr
-----	-----:	-----	-----:	-----	---	-----:	---	-----:
hellaswag	1	none	0	acc	↑	0.4577	±	0.0050
		none	0	acc_norm	↑	0.6168	±	0.0049

LM-Evaluation-Harness(MMLU)

- MMLU

```
lm_eval \  
--model hf \  
--model_args pretrained=meta-llama/Llama-3.2-1B-Instruct \  
--tasks mmlu \  
--device cuda:0 \  
--batch_size 2
```

LM-Evaluation-Harness(MMLU)

Tasks	Version	Filter	n-shot	Metric		Value		Stderr
-----	----	-----	-----	-----	---	-----	---	-----
mmlu	2	none		acc		0.4827	±	0.0041
- humanities	2	none	0	acc	↑	0.4587	±	0.0071
- formal_logic	1	none	0	acc	↑	0.3571	±	0.0429
- high_school_european_history	1	none	0	acc	↑	0.6606	±	0.0370
- high_school_us_history	1	none	0	acc	↑	0.5931	±	0.0345
- high_school_world_history	1	none	0	acc	↑	0.6456	±	0.0311
- international_law	1	none	0	acc	↑	0.6777	±	0.0427
- jurisprudence	1	none	0	acc	↑	0.5556	±	0.0480
- logical_fallacies	1	none	0	acc	↑	0.5276	±	0.0392
- moral_disputes	1	none	0	acc	↑	0.4653	±	0.0269
- moral_scenarios	1	none	0	acc	↑	0.3620	±	0.0161
- philosophy	1	none	0	acc	↑	0.5209	±	0.0284
- prehistory	1	none	0	acc	↑	0.5247	±	0.0278
- professional_law	1	none	0	acc	↑	0.3722	±	0.0123
- world_religions	1	none	0	acc	↑	0.6667	±	0.0362
- other	2	none	0	acc	↑	0.5494	±	0.0087
- business_ethics	1	none	0	acc	↑	0.4500	±	0.0500

LM-Evaluation-Harness(MMLU/GGUF)

- .gguf 평가 실행

```
pip install gguf
```

```
lm_eval --model hf --model_args \
        pretrained=/home/std/.lmstudio/models/lmstudio-community/Llama-3.2-1B-Instruct-GGUF,gguf
_file=Llama-3.2-1B-Instruct-Q4_K_M.gguf \
        --tasks mmlu \
        --device cuda:0 \
        --batch_size 8
```